Examination of radioactive radiation: Dosimetry, counting statistics and half-life determination

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Several dose rates and equivalent doses were measured for different exposure scenarios.The matching of a Poisson distribution could be demonstrated in two series of measurements, by performing a χ^2-Test. The half-life of the meta-stable state of Ba-137 could be determined to $t_{\frac{1}{2}} = (153,2 \pm 1,5)\,s$. The characteristics of a Geiger-Müller-tube was examined. The Threshold voltage and response of the counter tube used was determined.

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1 Introduction

Investigations in the field of radiation protection are not only very topical, but are essential for many areas of human technology. An unavoidable artificial radiation exposure arises from imaging procedures in medicine. Above all, the human body is exposed to high levels of radiation through the use of X-rays and computed tomography scan. According to Dr Amy Berrington de González and Prof Sarah Darby [3], 1.5% of the annual cancer cases in Germany can be traced back to radiation exposure from X-Ray-Radiography. In order to understand and improve radiation protection, various dose rates and radiation exposure were measure in a radiation experiment.

2 Theory

2.1 Radioactive decay

The time of decay of a radioactive material is purely non-deterministic. A statistical dependency can only be determined by considering a large number of atoms, the law of decay:

$$N(t) = N_0 e^{-\lambda(t-t_0)} \tag{1}$$

This formula can be used to describe how many radioactive particle $N(t)$ of the initial number N_0 are left after a certain time interval t_0. λ is a material constant.

Based on the law of decay, the activity A(t) of a radioactive substance can be derived:

$$A(t) = -\frac{dN(t)}{dt} \tag{2}$$

$$A(t) = -\lambda N(t) \tag{3}$$

The half-life $t_{1/2}$ describes the period of time in which half of a radioactive material has decayed.

$$t_{1/2} = \frac{ln2}{\lambda} \tag{4}$$

2.2 Gamma ray

γ-radiation are high-energy photons that occur after the spontaneous decay of a atomic nuclei as an accompaniment to alpha or beta decay. In general, photons with an energy of about 200 keV are referred to as γ-rays. In many, but not all, alpha and beta decays, the newly formed nucleus, also referred to as the daughter nucleus, is in a meta-stable excited state. During the transition to an stable state of lower energy, energy is released in the form of photons.

2.3 Math

To investigate the γ-radiation, the Poisson distribution was used. Samuel James Bader describes the usefulness of this distribution in the investigation of γ-radiation and explains why the Poisson distribution is suitable. [2]

2.4 Decay scheme of Cs-137

The cesium isotope Cs-137 has a half-life of 30 years and decays with the emission of beta radiation into Ba-137. Some of the daughter nucleus formed are ion a meta-stable excited state. With the emission of gamma ray, with an energy of 0.661 MeV, this state changes into the basic state of Ba-137. The decay scheme is shown in Figure 1.

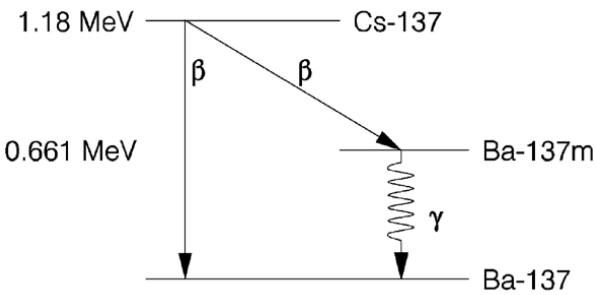

Figure 1: Decay scheme of Cs-137 with its stable and meta-stable state-

3 Experiment

3.1 Activity of the Cs-137 probe

The used Cs-137 probe is from August 1978. The manufacturer stated that is had an activity of 3,7 MBq. With the half-life of 30,2 years the activity of the probe could be calculated to

$$A_{\mathrm{Cs,today}} = 1,408\,\mathrm{MBq} \tag{5}$$

3.2 Absorbed equivalent dose

In this chapter we want to determine the absorbed equivalent doses for different exposure scenarios. The doses were measured with the dose rate meters NEVA and RADEX RD 1503.

3.2.1 Local dose in the laboratory

The dose rate meter RADEX RD 1503 was put on a wall in the laboratory. The dose rate was measured every 15min. With the average rate we calculate the equivalent dose for the time we stayed in the laboratory. The average dose rate of the time we spent in the laboratory and its standard error is $\dot{D} = (0,1108 \pm 0,0065)\,\mu\mathrm{Sv/h}$. We spent about 3 hours in total in the laboratory. This leads to an absorbed equivalent dose of

$$D = (0,332 \pm 0,020)\,\mu\mathrm{Sv} \tag{6}$$

If you were to calculate the dose for a hole year, that is 8760 hours, you would get a dose of

$$D = (0,971 \pm 0,057)\,\mathrm{mSv} \tag{7}$$

3.2.2 Equivalent dose from the Cs-137 probe

The dose rate from a decaying Cs-137 probe was measured with the NEVA dose rate meter. The probe is encased in a lead cube. On the top it has a small opening, which is covered with a plug. By removing the plug, radiation can leave the cube. With the plug removed, we measured the dose rate right in front of the hole and in a distance of approximately 1m. With this we calculate the equivalent dose we get from standing in front of the probe.

The dose rate measured right in front of the probe opening is $(5,8 \pm 0,1)\,\mu\mathrm{Sv/h}$. Without the data-sheet of the dose meter being available, we assume an error in the last digit. In a time span of 4 hours that would be an absorbed dose of

$$D = (23,2 \pm 0,4)\,\mu\mathrm{Sv} \tag{8}$$

The dose rate that was 1 meter above the opening showed a dose rate of $\dot{D} = (0,6 \pm 0,1)\,\mu\text{Sv/h}$. From this rate we subtract the background rate from the laboratory and get

$$\dot{D} = (0,49 \pm 0,14)\,\mu\text{Sv/h} \tag{9}$$

With this we can calculate the time we are allowed to work there without overstepping the limit of 5 mSv to

$$t = \frac{5\,\text{mSv}}{0,49\,\mu\text{Sv/h}} = 10204,1\text{h} \tag{10}$$

This is more hours than a year has. It is safe to say the one could work uninterrupted there for a whole year without overstepping the limit. During a time span of 4 hours one would only absorb 0,0392% of the given limit.

3.3 Measuring the counting tube characteristics

The characteristics of a Geiger-Müller-counter was measured. For this we put the counter in front of the opening of the Cs-137 probe. We then measured the number of counts for 60 seconds for different voltages of the counter. With this we chose the voltage at which the counter will be operated. The results can be seen in Figure 2. One can see that the threshold voltage, the voltage at which the counter starts counting properly, is at about (510 ± 2) V. After this, there is a plateau where the number of counts are nearly constant. It is best to use the Geiger-Müller-counter in this area. We chose a voltage that is in the middle of this plateau. From the graph we chose 560 V for our working voltage. This voltage is used for every following experiment using this counting-tube.

3.4 Background radiation

With all radiation sources closed and put aside, we measured the number of decays from the background for about 5 minutes. The counter is then put in a lead surrounding, where the number of decays was also measured for 5 minutes.

In a time span of t=300,7 s we counted N=156 events.

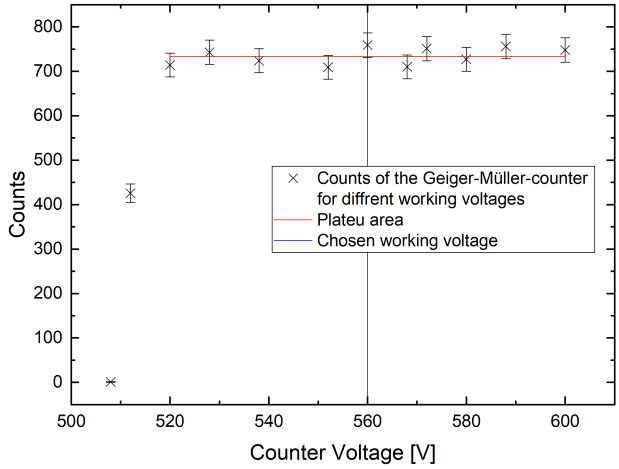

Figure 2: Counting-tube characteristics. The number of counts is measured for different voltages of the counting-tube. The tube only starts counting at a certain threshold voltage. This voltage can be read at about (510±2) V. After a rapid incline in counts, the number of counts stays about the same. The working voltage should ideally be in the middle of this plateau. For this we chose the working voltage to 560 V.

With this we get a background of

$$A_{0,\text{room}} = (0,519 \pm 0,031)\,\text{Bq} \tag{11}$$

The experiment was repeated with the counter surrounded by a lead structure. In a time span of t=297,75 s we counted N=86 events. With this we get a background of

$$A_{0,\text{lead}} = (0,289 \pm 0,031)\,\text{Bq} \tag{12}$$

In the lead structure the background radiation was reduced about 44,3%. With the measured background, data from the other experiments can be corrected accordingly.

3.5 Counting statistics

A orange tile of uranium-oxide was used as a probe, to test counting statistics. The tile and counting-tube were both encased in a lead structure. Two measurement series were carried out. In both measurements we counted about 1000 times. In one measurement the gate time was set to 1 s in the other to 10 s. For both series a χ^2-Test was made, to determine if they follow a Poisson-distribution. In Figure 3 the bar graph of the measuring series with a gate time of 1 second is

shown. A Poisson-curve was added to the graph. From the statistic we get a mean value and a standard deviation of

$$\overline{x} = 0,78 \tag{13}$$

$$\sigma = 0,88 \tag{14}$$

With this we do a χ^2-Test, to confirm a Poisson distribution. We test with a significance constant of 5%. The event for 4 and 5 counts was combined, because there were less than 5 events in the class. We get a total $\chi^2 = 0.75$. To reject the assumption we need a χ^2 of at least 9,49. This is not the chase, so we can safley say that the number of events follow a Poisson distribution. The possibility that a χ^2 of at least this value occurs is 94,5%.

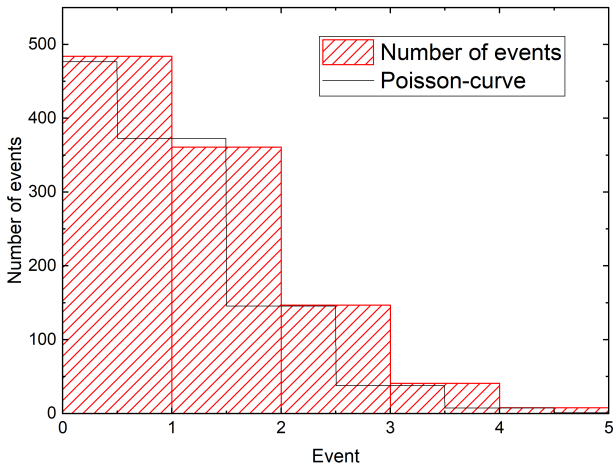

Figure 3: The counting statistics of an uranium-oxid plated tile. The gate time was set to 1 second. A Poisson-curve was added to the graph.

In Figure 4 the bar graph of the measuring series with a gate time of 10 seconds is shown. A Normal-curve was added to the graph. The standard deviation was determined graphically with a cumulative plot that can be seen in the appendix in Figure 6.

$$\sigma = 2,8 \pm 0,1 \tag{15}$$

This fits approximately to the expected value of $\sigma = \sqrt{\overline{N}} = 2,82$. To further examine the normal-distribution a χ^2-Test was performed. Different classes were put together to have at least 5 counts per class. With a significance constant of 5% the critical value is χ^2=25. We get χ^2=26,87. This exceeds the limit, so we have to deny the assumption that the data follows

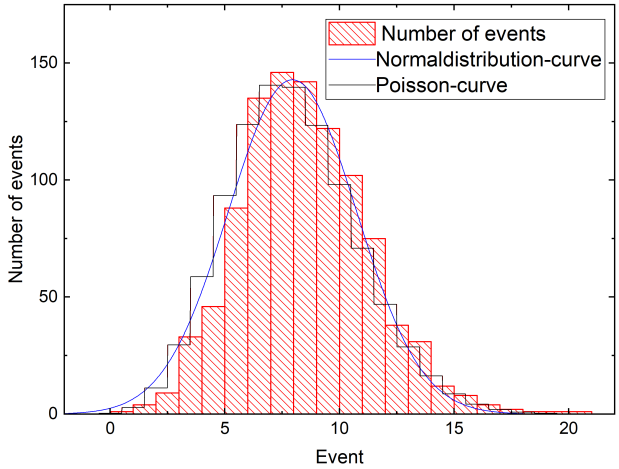

Figure 4: The counting statistics of an uranium-oxid plated tile. The gate time was set to 10 seconds. A Normal-curve was added to the graph.

a normal-distribution. In only 2,98% of the cases we expect a deviation at least this high.

For this series a χ^2-Test, to confirm a Poisson distribution, was also carried out. Like before, a few classes were put together to get at least 5 events per class. The resulting χ^2=9,9. With a significance constant of 5% the χ^2-value should not exceed χ^2=25 to confirm Poisson-distribution. This is not the case, so we can again confirm the Poisson-distribution. A deviation of this kind occurs in 82,3% of the cases.

3.6 Determination of the half-life of Ba-137m

To determine the half-life of Ba-137m the activity of the probe was measured over time. Ba-137m was extracted from the Cs-probe by rinsing it out with hydrochloric acid. With this method we can assure that we only have Ba-137m in our probe. Because of the low half-life the probe is put immediately in front of the Geiger-Müller-counter to begin the measurement.

The activity of Ba-137m is measured for about 25 minutes. The number of counts was adjusted for the background radiation and plotted over time. The graph can be seen in Figure 5. An exponential curve was fitted on the curve. With the resulting decay-constant, we calculate the half-life to

$$t_{\frac{1}{2}} = (153,2 \pm 1,5)\,s \tag{16}$$

This fits, within its error bars, with the literature value

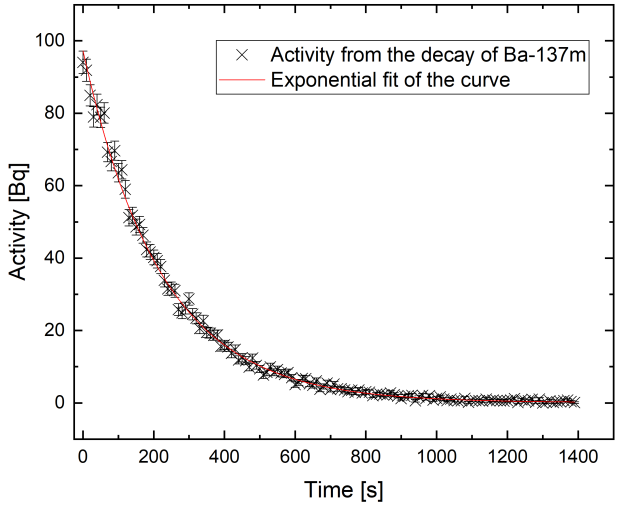

Figure 5: Measured activity over time of the decaying of Ba-137m. A exponential curve was fitted to the data, to determine the half-life to $t_{\frac{1}{2}} = (153,2 \pm 1,5)$ s,

of $t_{\frac{1}{2}} = 153,12$ s [1].

3.7 Sensitivity of the counting tube

The sensitivity of the counting tube is measured. For this the Cs-137 probe is put out from its lead housing. The probe can now radiate in all directions. The counting-tube is put near the probe at about 60 cm distance. A counting measurement is made for 60 seconds. With the number of counts form the tube, minus the background radiation, and the known activity of the Cs-137 probe, we can calculate the sensitivity of the counting-tube.

The distance between probe and tube was r=(58,5±0,5)cm. At this distance we measured 201 decays. Without the background we get an activity of

$$\mathrm{A}_{\mathrm{measured}} = (2,83 \pm 0,23)\,\mathrm{Bq} \tag{17}$$

The effective surface of the counting-tube is A_{tube}=11 cm^2. With this we can the calculate the solid angle that the probe sees. We get for the expected activity

$$\mathrm{A}_{\mathrm{expected}} = \mathrm{A}_{\mathrm{Cs}} \frac{A_{\mathrm{tube}}}{4\pi r^2} = (360,1 \pm 6,2)\,\mathrm{Bq} \tag{18}$$

Where A_{Cs} is the calculated activity of the Cs-probe. With this we get a sensitivity of

$$\frac{\mathrm{A}_{\mathrm{measured}}}{\mathrm{A}_{\mathrm{expected}}} = (0,786 \pm 0,065)\,\% \tag{19}$$

4 Summary

With the help of a dose rate meter, we could determine the radiation exposure for different scenarios. By performing a χ^2-Test, we could confirm the Poisson-distribution for two measurement series of the decay of an uranium-oxid plated tile. Through examining the characteristics of a Geiger-Müller-tube we could determine the threshold voltage and the working voltage. The half-life of Ba-137m could be determined to $t_{\frac{1}{2}} = (153,2 \pm 1,5)\,s$. The sensitivity of our counting-tube was calculated to (0,786±0,065)%.

5 Appendix

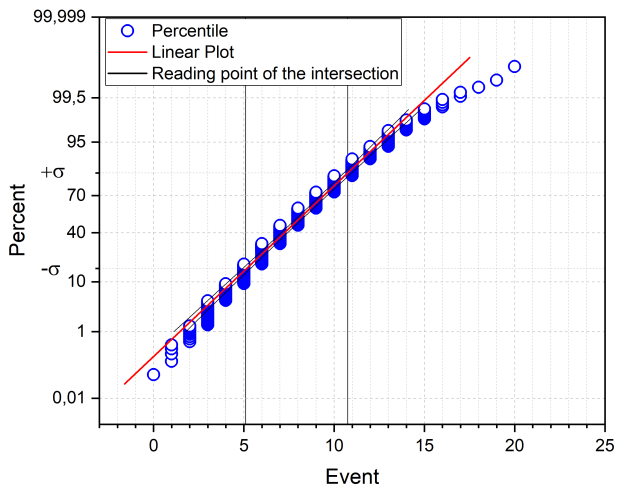

Figure 6: Cumulative plot of the measurement series with a gate-time of 10s. To determine the standard deviation a line was fitted to the graph. At the intersection between the fitted line and percentage of the standard deviation the value of the deviation could be read. With this the standard deviation was determined to $\sigma = 2,8 \pm 0,1$.

References

[1] Half-live of ba-137m. National Nuclear Data Center, Brookhaven National Laboratory based on ENSDF and the Nuclear Wallet Cards

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[2] KAYO ARIMA and JOHN WOOLEY. 1. overview and scientific context. *Computational Methods for Understanding Bacterial and Archaeal Genomes*, 7:345, 2008.

[3] Amy Berrington de Gonzalez and Sarah Darby. Risk of cancer from diagnostic x-rays: estimates for the uk and 14 other countries. *The lancet*, 363(9406):345–351, 2004.